Anticipating the Bid: Bilevel Optimization for Pediatric Vaccine Tenders

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Abstract

NU: In this study, we explore .

Keywords: Bilevel Optimization, Stackleberg game, Vaccine Tender

1 Introduction

2 Literature Review

3 Modeling

3.1 Nomenclature

Symbols marked † are introduced for the bilevel model; all others follow the manuscript.

Sets

Symbol	Description
$T = \{1, \ldots, N\}$	Planning periods (years), horizon length N
A	Antigens
V	Vaccines

P	Pre-qualified producers
M	All-unit discount brackets
$= \{(t,\tau) : t \le \tau \le \min(t+\Delta-1, N)\}$	Feasible tender windows
$A_p \subseteq A,\ V_p \subseteq V$	Antigens and vaccines producer p supplies
$P_a,\ P_v \subseteq P$	Producers able to supply antigen a, vaccine v
$A_v \subseteq A,\ V_a \subseteq A$	Antigens in vaccine v; vaccines containing antigen a
K	Price ladder levels
J	Award-share levels
$(l) = \{(t,\tau) \in: t \le l \le \tau\}$	Tender windows active in period l

Parameters

Symbol	Description
δ_t	Time-discount factor for period t
g_t	Tender set-up cost
$\bar{r}_{vt}$	Reference (reservation proxy) price per dose of vaccine v
r_{vpt}	Reservation price of producer p for vaccine v
ζ_{vm}	Discount rate of bracket m, increasing in m
$\phi^L_{vm},\ \phi^U_{vm}$	Lower and upper quantity bounds of bracket m
β	Penalty per missed dose
h_v	Holding cost as a fraction of price
d_{at}	Demand for antigen a in period t
$\hat{I}_{v0}$	Initial inventory
$\bar{s}_p$	Base capacity of producer p, shared across V_p
s_{pt}	Available capacity of producer p in period t
κ	Allowable capacity expansion rate per unit of L
γ_p	Cost per unit of capacity expansion
$f_{vpt\tau}$	Production set-up cost for a commitment over (t,τ)
l_{vp}	Required return margin in the ROI constraint
Δ	Maximum tender length

λ_k	Price multiplier of ladder level k
$\rho_{vtk} = \lambda_k \bar{r}_{vt}$	Price per dose at level k
u_j	Award share of level j, in $[0, 1]$
$\hat{d}_{vt}$	Vaccine-level demand forecast
$\hat{D}_{vt\tau} = \sum_{l=t}^{\tau} \hat{d}_{vl}$	Tender volume of vaccine v over (t, τ)
c_{vp}	Per-dose production cost (synthetic)
$\bar{L}_p$	Upper bound on capacity expansion units

Decision variables

Symbol	Domain	Description	Bilevel owner
$F_{at\tau}$	$\{0, 1\}$	Tender for antigen a opened over (t, τ)	Leader
Π_{vtk}	$\{0, 1\}$	Price level k for vaccine v in tenders starting at t	Leader
$\Omega_{vpt\tau j}$	$\{0, 1\}$	Award-share level j for producer p, vaccine v, window (t, τ)	Leader
I_{vt}	≥ 0	Inventory of vaccine v at end of t	Leader
V_{vt}	≥ 0	Doses of vaccine v administered in t	Leader
S_{at}	≥ 0	Cumulative missed doses of antigen a	Leader
$W_{pt\tau}$	$\{0, 1\}$	Producer p commits to window (t, τ)	Follower p
Y_{pt}	$\{0, 1\}$	Producer p produces in t	Follower p
Z_{vptm}	$\{0, 1\}$	Bracket m active for (v, p, t)	Follower p
L_{pt}	$\mathbb{Z}_+$	Capacity expansion units	Follower p
$Q_{vpt\tau m}$	≥ 0	Committed quantity in bracket m	Follower p
X_{vpt}	≥ 0	Doses delivered in t (single-level model)	Follower p
$X_{vpt\tau l}$	≥ 0	Doses delivered in l against window (t, τ)	Follower p
$\Gamma_{vpt\tau mk}$	≥ 0	Part of $Q_{vpt\tau m}$ priced at level k	Follower p
$\omega_{pt\tau k},\ \eta_{ptl}$	≥ 0	Linearization variables (Section 5)	Follower p

4 Single-level model (current)

In the single-level model a coordinating entity chooses every variable. The UNICEF–Gavi objective (UG) minimizes discounted tendering, procurement, missed-dose and holding costs. The producer profit objective (MP) is stated after it because it becomes the lower level of the bilevel model.

Model UG

$$\min \sum_{t\in T}\delta_t\left(\sum_{a\in A}\sum_{\tau:(t,\tau)\in} g_t F_{at\tau}+\sum_{v\in V}\sum_{p\in P_v}\sum_{m\in M}\sum_{\tau:(t,\tau)\in}\bar{r}_{vt}(1-\zeta_{vm})Q_{vpt\tau m}+\sum_{a\in A}\beta S_{at}+\sum_{v\in V}h_v\bar{r}_{vt}I_{vt}\right) \tag{1}$$

subject to

$$(\tau-t+1)F_{at\tau}\le\sum_{p\in P_a}\sum_{l=t}^{\tau}Y_{pl}\qquad\forall a\in A,\ (t,\tau)\in \tag{2}$$

$$\sum_{\tau:(t,\tau)\in}F_{at\tau}\le 1\qquad\forall a\in A,\ t\in T \tag{3}$$

$$\sum_{(l,k)\in:\ l\le t\le k}F_{alk}\ge 1\qquad\forall a\in A,\ t\in T \tag{4}$$

$$\sum_{a\in A_p}F_{at\tau}\ge W_{pt\tau}\qquad\forall p\in P,\ (t,\tau)\in \tag{5}$$

$$F_{at\tau}\le\sum_{p\in P_a}W_{pt\tau}\qquad\forall a\in A,\ (t,\tau)\in \tag{6}$$

$$\sum_{v\in V_p}\sum_{m\in M}Q_{vpt\tau m}\le W_{pt\tau}\left(\sum_{l=t}^{\tau}\bar{s}_p+\sum_{k=1}^{t}\kappa(\tau-t+1)\bar{s}_pL_{pk}\right)\qquad\forall p\in P,\ (t,\tau)\in \tag{7}$$

$$\sum_{v\in V_p}\sum_{m\in M}\bar{r}_{vt}(1-\zeta_{vm})Q_{vpt\tau m}\ge\sum_{v\in V_p}(1+l_{vp})f_{vpt\tau}W_{pt\tau}\qquad\forall p\in P,\ (t,\tau)\in \tag{8}$$

$$\phi^L_{vm}Z_{vptm}\le\sum_{\tau:(t,\tau)\in}Q_{vpt\tau m}\qquad\forall v\in V,\ p\in P_v,\ t\in T,\ m\in M \tag{9}$$

$$\phi^U_{vm} Z_{vptm} \geq \sum_{\tau:(t,\tau)\in} Q_{vpt\tau m} \qquad \forall v \in V,\ p \in P_v,\ t \in T,\ m \in M \tag{10}$$

$$\sum_{m\in M} Z_{vptm} = 1 \qquad \forall v \in V,\ p \in P_v,\ t \in T \tag{11}$$

$$\sum_{m\in M} Q_{vpt\tau m} \geq \sum_{l=t}^{\tau} X_{vpl} \qquad \forall v \in V,\ p \in P_v,\ (t,\tau) \in \tag{12}$$

$$\sum_{v\in V_p} X_{vpt} \leq Y_{pt}\left(s_{pt} + \sum_{l=1}^{t} \kappa \bar{s}_p L_{pl}\right) \qquad \forall p \in P,\ t \in T \tag{13}$$

$$I_{v,t-1} + \sum_{p\in P_v} X_{vpt} = V_{vt} + I_{vt} \qquad \forall v \in V,\ t \in T \tag{14}$$

$$d_{at} - \sum_{v\in V_a} V_{vt} + S_{a,t-1} \leq S_{at} \qquad \forall a \in A,\ t \in T \tag{15}$$

$$I_{v0} = \hat{I}_{v0} \qquad \forall v \in V \tag{16}$$

$$F_{at\tau},\ Y_{pt},\ W_{pt\tau},\ Z_{vptm} \in \{0,1\}, \quad L_{pt} \in \mathbb{Z}_+ \tag{17}$$

$$Q_{vpt\tau m},\ X_{vpt},\ I_{vt},\ S_{at},\ V_{vt} \geq 0 \tag{18}$$

The manuscript's social-benefit model (SB), equation (19), is outside the scope of this document.

Model MP (lower level of the bilevel model)

MP maximizes producer net revenue subject to (2)–(18), written as a minimization:

$$\min \sum_{t\in T} \delta_t \left(- \sum_{v\in V} \sum_{p\in P_v} \sum_{m\in M} \sum_{\tau:(t,\tau)\in} \bar{r}_{vt}(1-\zeta_{vm}) Q_{vpt\tau m} + \sum_{p\in P} \gamma_p L_{pt} + \sum_{p\in P} \sum_{v\in V_p} \sum_{\tau:(t,\tau)\in} f_{vpt\tau} W_{pt\tau} \right) \tag{20}$$

5 Linearization

Constraints (7) and (13) multiply a binary by the integer L. With $L_{pt} \leq \bar{L}_p$, each product is replaced exactly by a continuous auxiliary variable and four McCormick inequalities (Glover, 1975; McCormick, 1976).

For (7), let $\omega_{pt\tau k} = W_{pt\tau} L_{pk}$:

$$\begin{aligned} &\omega_{pt\tau k} \leq \bar{L}_p W_{pt\tau}, \\ &\omega_{pt\tau k} \leq L_{pk}, \\ &\omega_{pt\tau k} \geq L_{pk} - \bar{L}_p(1 - W_{pt\tau}), \\ &\omega_{pt\tau k} \geq 0 \end{aligned} \qquad \forall p,\ (t,\tau) \in,\ k \leq t \tag{21}$$

$$\sum_{v \in V_p} \sum_{m \in M} Q_{vpt\tau m} \leq (\tau - t + 1)\bar{s}_p W_{pt\tau} + \kappa(\tau - t + 1)\bar{s}_p \sum_{k=1}^{t} \omega_{pt\tau k} \qquad \forall p,\ (t,\tau) \in \tag{22}$$

For (13), let $\eta_{ptl} = Y_{pt} L_{pl}$:

$$\begin{aligned} &\eta_{ptl} \leq \bar{L}_p Y_{pt}, \\ &\eta_{ptl} \leq L_{pl}, \\ &\eta_{ptl} \geq L_{pl} - \bar{L}_p(1 - Y_{pt}), \\ &\eta_{ptl} \geq 0 \end{aligned} \qquad \forall p,\ t,\ l \leq t \tag{23}$$

$$\sum_{v \in V_p} X_{vpt} \leq s_{pt} Y_{pt} + \kappa \bar{s}_p \sum_{l=1}^{t} \eta_{ptl} \qquad \forall p,\ t \tag{24}$$

Constraint (22) replaces (7) and (24) replaces (13) in every model below. Both products involve only producer variables, so in the bilevel model the linearization sits entirely inside each follower.

5.1 Bilevel model

The bilevel model places UG over MP. The coordinating entity (leader, objective (1)) announces the tender calendar, per-dose prices and award shares. Each producer (follower, objective (20) restricted to its own variables) then chooses its participation, commitments, capacity and deliveries to maximize its own profit. Let $x = (F, \Pi, \Omega)$ denote the linking decisions and $y_p = (W, Y, Z, L, Q, X, \Gamma, \omega, \eta)$ the variables of producer p.

Assumptions

1. Open-loop commitment: the leader fixes the full calendar and terms at the start of the horizon, consistent with published multi-year tender calendars.

2. Producers do not cooperate and share no constraint; they interact only through the leader's decisions.

3. Complete information: the leader knows each producer's costs and capacities (synthetic in the experiments).

4. Optimistic convention: among a producer's optimal responses, the one best for the leader is selected.

5. Prices are chosen from a finite ladder, and discount brackets (ζ, ϕ) are announced in advance as a menu from which producers self-select.

Upper level (leader)

$$\min_{x,I,V,S} \sum_{t\in T}\delta_t\Bigg(\sum_{a\in A}\sum_{\tau:(t,\tau)\in} g_t F_{at\tau} + \sum_{v\in V}\sum_{p\in P_v}\sum_{\tau:(t,\tau)\in}\sum_{m\in M}\sum_{k\in K}\rho_{vtk}(1-\zeta_{vm})\Gamma_{vpt\tau mk} + \sum_{a\in A}\beta S_{at} + \sum_{v\in V} h_v \bar{r}_{vt} I_{vt}\Bigg) \tag{25}$$

subject to (3), (4), (15), (16) and

$$\sum_{k\in K}\Pi_{vtk} \le \sum_{a\in A_v}\sum_{\tau:(t,\tau)\in} F_{at\tau} \qquad \forall v\in V,\ t\in T \tag{26}$$

$$\sum_{k\in K}\Pi_{vtk} \le 1 \qquad \forall v\in V,\ t\in T \tag{27}$$

$$\sum_{j\in J}\Omega_{vpt\tau j} \le \sum_{a\in A_v} F_{at\tau} \qquad \forall v\in V,\ p\in P_v,\ (t,\tau)\in \tag{28}$$

$$\sum_{j\in J}\Omega_{vpt\tau j} \le 1 \qquad \forall v\in V,\ p\in P_v,\ (t,\tau)\in \tag{29}$$

$$I_{v,t-1} + \sum_{p \in P_v} \sum_{(t',\tau) \in (t)} X_{vpt'\tau t} = V_{vt} + I_{vt} \qquad \forall v \in V,\ t \in T \tag{30}$$

$$F_{at\tau},\ \Pi_{vtk},\ \Omega_{vpt\tau j} \in \{0,1\}; \quad I_{vt},\ V_{vt},\ S_{at} \geq 0 \tag{31}$$

$$y_p \in \mathcal{R}_p(x) \qquad \forall p \in P \tag{32}$$

Lower level (follower p)

$$\begin{aligned} \mathcal{R}_p(x) = \arg\max_{y_p} \sum_{t \in T} \delta_t \Bigg(& \sum_{v \in V_p} \sum_{\tau:(t,\tau) \in} \sum_{m \in M} \sum_{k \in K} \rho_{vtk}(1 - \zeta_{vm}) \Gamma_{vpt\tau mk} \\ & - \gamma_p L_{pt} \\ & - \sum_{v \in V_p} \sum_{\tau:(t,\tau) \in} f_{vpt\tau} W_{pt\tau} \\ & - \sum_{v \in V_p} \sum_{(t',\tau) \in (t)} c_{vp} X_{vpt'\tau t} \Bigg) \end{aligned} \tag{33}$$

Objective (33) is the MP objective for producer p alone, with revenue priced at the leader's chosen level and one added term, the production cost c_{vp}. Setting $c_{vp} = 0$ recovers (20) exactly. It is subject to (21)–(23), (9)–(11) for $v \in V_p$, and

$$W_{pt\tau} \leq \sum_{a \in A_p} F_{at\tau} \qquad \forall (t,\tau) \in \tag{34}$$

$$\sum_{m \in M} Q_{vpt\tau m} \leq \hat{D}_{vt\tau} \sum_{j \in J} u_j \Omega_{vpt\tau j} \qquad \forall v \in V_p,\ (t,\tau) \in \tag{35}$$

$$\sum_{m \in M} Q_{vpt\tau m} \leq \hat{D}_{vt\tau} W_{pt\tau} \qquad \forall v \in V_p,\ (t,\tau) \in \tag{36}$$

$$\sum_{k \in K} \Gamma_{vpt\tau mk} = Q_{vpt\tau m} \qquad \forall v \in V_p,\ (t,\tau) \in,\ m \in M \tag{37}$$

$$\Gamma_{vpt\tau mk} \leq \hat{D}_{vt\tau} \Pi_{vtk} \qquad \forall v \in V_p,\ (t,\tau) \in,\ m \in M,\ k \in K \tag{38}$$

$$\sum_{l=t}^{\tau} X_{vpt\tau l} = \sum_{m \in M} Q_{vpt\tau m} \qquad \forall v \in V_p,\ (t,\tau) \in \tag{39}$$

$$\sum_{v \in V_p} \sum_{(t',\tau) \in (t)} X_{vpt'\tau t} \leq s_{pt} Y_{pt} + \kappa \bar{s}_p \sum_{l=1}^{t} \eta_{ptl} \qquad \forall t \in T \tag{40}$$

$$W_{pt\tau}, Y_{pt}, Z_{vptm} \in \{0,1\}; \quad L_{pt} \in \{0, \ldots, \bar{L}_p\}; \quad Q, X, \Gamma, \omega, \eta \geq 0 \tag{41}$$

Remarks. Constraints (37)–(38) price each commitment exactly at the chosen ladder level without a bilinear term, so every follower has constant objective coefficients and the leader's variables enter only the right-hand sides, the setting assumed by exact MIBLP methods. Setting $y_p = 0$ is feasible for every x and yields zero profit, so individual rationality holds without the ROI constraint (8). Constraints (30) and (15) contain follower deliveries but are always satisfiable through V, I and S, so the only coupling between levels is through the leader's objective.

5.2 Equivalent forms and bounds

Single-follower equivalent. Let $\mathcal{Y}_p(x)$ be the feasible set of follower p, defined by (9)–(11), (21)–(23) and (34)–(41), and $\pi_p(x, y_p)$ its objective (33). The sets $\mathcal{Y}_p(x)$ share no constraint and the objectives are separable, so

$$\prod_{p \in P} \mathcal{R}_p(x) = \arg\max \left\{ \sum_{p \in P} \pi_p(x, y_p) : y_p \in \mathcal{Y}_p(x) \ \ \forall p \in P \right\} \tag{42}$$

The model can therefore be passed to a single-follower MIBLP solver such as MibS (Tahernejad, Ralphs & DeNegre, 2020) with the summed producer profit as the lower-level objective, while decomposition methods can still solve each producer's problem separately.

Value-function reformulation. With $\varphi_p(x) = \max\{\pi_p(x, y_p) : y_p \in \mathcal{Y}_p(x)\}$, the bilevel model is equivalent to

$$
\begin{aligned}
&\min\ (25)\\
&\text{s.t.}\quad (3),(4),(15),(16),(26)\text{–}(31),\\
&\qquad y_p \in \mathcal{Y}_p(x),\\
&\qquad \pi_p(x,y_p) \geq \varphi_p(x) \qquad \forall p \in P
\end{aligned}
\tag{43}
$$

The last constraint is nonconvex and defined implicitly; reformulation-and-decomposition algorithms (Zeng & An, 2014; Yue, Gao, Zeng & You, 2019) approximate φ_p from below by enumerating follower responses and converge finitely when the linking variables x are integer, which holds here.

High-point relaxation. Dropping $\pi_p \geq \varphi_p$ from (43) gives the high-point relaxation (HPR), a single MILP:

$$
\begin{aligned}
v^{HPR} = \min\ &(25)\\
\text{s.t.}\quad &(3),(4),(15),(16),(26)\text{–}(31),\\
&y_p \in \mathcal{Y}_p(x) \qquad \forall p \in P
\end{aligned}
\tag{44}
$$

For the bilevel optimal value v^{BL}, $v^{HPR} \leq v^{BL}$. The cost of producer autonomy is defined as

$$
\text{CPA} = \frac{v^{BL} - v^{HPR}}{v^{HPR}} \tag{45}
$$

With one price level ($|K| = 1$, $\lambda_1 = 1$), one award level ($u = 1$), and constraints (2), (6) and (8) restored, the HPR reduces to model UG apart from the tender index on deliveries. UG with (8) is therefore the manuscript's approximation of producer rationality, and its schedule can be evaluated against producers' true best responses.

6 Constraint map

Single-level	Bilevel	Change

Objective (1)	Leader objective (25)	Fixed price $\bar{r}_{vt}$ replaced by chosen ladder price through Γ
Objective (20)	Follower objective (33)	Split per producer; adds production cost c_{vp}
(2) tender requires production	Removed	Producers' Y is their own choice; an unserved tender appears as missed doses
(3), (4) tender structure	Leader	Unchanged
(5) commit only to open tenders	Follower (34)	F is a parameter to the follower
(6) tender requires a committed producer	Removed	Same reason as (2)
(7) commitment capacity	Follower (21)–(22)	Linearized
(8) ROI	Removed	Replaced by follower optimality; zero response always feasible
(9)–(11) discount brackets	Follower	Producers self-select brackets
(12) commitment covers deliveries	Follower (39)	Equality on tender-indexed deliveries
(13) per-period capacity	Follower (23), (40)	Linearized; deliveries summed over active windows
(14) inventory balance	Leader (30)	Deliveries come from followers
(15), (16) missed doses, initial stock	Leader	Unchanged
New	Leader (26)–(29)	Price ladder and award shares
New	Follower (35)–(38)	Award ceiling, commitment link, price split

7 Analysis and Discussion of Results

8 Conclusion

Acknowledgements

Funding: This work was partially supported by The Bill & Melinda Gates Foundation, Seattle, WA [grant number INV-044813].

This research is supported by the Arkansas High-Performance Computing Center, which is funded through multiple National Science Foundation grants and the Arkansas Economic Development Commission.

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